



# Siriyak CR

AI-AUGMENTED BIOSTATISTICIAN & CLINICAL DATA AUTOMATION SPECIALIST

Bangalore BTM

Email: siriyak2016@gmail.com

Clinical Data Analyst-Statistician with 8+ years of experience (6+ years in Clinical Domain). Expert in R, SAS, and LLM/Agentic AI automation for clinical trials and real-world evidence (RWE). Proficient in CDISC standards (SDTM/ADaM) and end-to-end statistical programming. Experience in SAP development and TLF generation. Specialized in oncology endpoints and survival analysis (OS/PFS).

## Skills

### Core Expertise

8+ Years

CLINICAL BIOSTATISTICS | REAL-WORLD EVIDENCE (RWE) | STATISTICAL PROGRAMMING | DATA ANALYTICS | CLINICAL TRIAL DESIGN

### LLM & Agentic AI (Core Expertise)

80%+ Automation

AGENT-BASED WORKFLOW DESIGN ACHIEVING 80%+ TASK AUTOMATION ACROSS CLINICAL BIOSTATISTICS WORKFLOWS INCLUDING STATISTICAL ANALYSIS PLANS (SAPs), TABLES/LISTINGS/FIGURES (TLFs), RECIST LOGIC IMPLEMENTATION, AND REGULATORY DOCUMENTATION. PLATFORMS: GEMINI (ANTIGRAVITY), CLAUDE HAIKU/SONNET, AMAZON Q, GPT-4. ADVANCED PROMPT ENGINEERING, MULTI-AGENT ORCHESTRATION, LLM-ASSISTED CODE GENERATION (R/SAS/SQL/PYTHON), AUTONOMOUS VALIDATION PIPELINES, AND END-TO-END CLINICAL DATA WORKFLOW AUTOMATION

### Statistical Methods & Analytics

Expert Level

SURVIVAL ANALYSIS (KAPLAN-MEIER, COX PROPORTIONAL HAZARDS, LOG-RANK TESTS, RWPFS, RWOS, RWDOT, RWTTD, TTNT, DOR, HAZARD MODELING) | BAYESIAN STATISTICS (MCMC, COMPUTATIONAL STATISTICS, BAYESIAN MODELING FOR RISK-BASED MONITORING) | MACHINE LEARNING (RANDOM FOREST, XGBOOST, SVM, KNN, NEURAL NETWORKS, CLASSIFICATION, DECISION TREES, PATTERN RECOGNITION) | ADVANCED ANALYTICS (DOE, FACTORIAL DESIGN, PCA, FACTOR ANALYSIS, DISCRIMINANT ANALYSIS, MDS, PARTIAL LEAST SQUARES, CLUSTER ANALYSIS, MULTIVARIATE ANALYSIS) | PREDICTIVE MODELING | REGRESSION ANALYSIS | STATISTICAL QUALITY CONTROL | EXACT TESTS | INFERENCE STATISTICS

### Therapeutic Areas

6+ Years

ONCOLOGY | NEUROSCIENCE | AUTOIMMUNE

### R Shiny Development

Production Apps

PRODUCTION DASHBOARD DEVELOPMENT | CLINICAL TRIAL MONITORING | RISK ANALYSIS | REAL-TIME VISUALIZATION | SHINYDASHBOARD | SHINYAPPS.IO | RSTUDIO CONNECT

### Programming & Tools

8+ Years

R | SAS (CLINICAL PROGRAMMING) | PYTHON | SQL | GIT | RMARKDOWN | BASH | RESTFUL APIS | PLUMBER

### Clinical Standards

ICH-GCP Certified

CDISC (SDTM/ADaM) | FDA SUBMISSIONS | GxP COMPLIANCE | ICH-GCP | SAP DEVELOPMENT | TLF GENERATION | REGULATORY REPORTING

### Data Platforms

Multi-Platform

VEEVA EDC | JREVIEW | MEDIATA RAVE | SPOTFIRE | POWER BI | POSTGRES SQL | DBEAVER | EXCEL

### Leadership & Tools

Team Lead

TEAM LEAD | AGILE/SCRUM | JIRA | CONFLUENCE | TRAINING & MENTORING | CROSS-FUNCTIONAL COLLABORATION

## Education

### University of Mysore

Mysore

M.S.C IN STATISTICS

### Yuvaraj College ,University of Mysore

Mysore

B.S.C IN MATHEMATICS, STATISTICS, COMPUTER SCIENCE

## Biostatistician / Lead Clinical Database Programmer

Mumbai

April 2024 -> present

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- LANGUAGES/TOOLS USED – Veeva, CQL, R, PostgreSQL
- Utilized Veeva for listing and edit checks, study deployment.
- Developed and implemented R scripts for data analysis, reporting, and data quality checks.
- Managed PostgreSQL databases for JReview listings and reports.
- Ensured data quality and integrity through rigorous validation processes.
- Collaborated with cross-functional teams to deliver comprehensive clinical data solutions.
- Provided training and support to team members on data management and analysis tools.
- Analyzed clinical trial datasets with R and SAS to ensure regulatory compliance.
- Created statistical analysis plans (SAP) and supported SDTM/ADaM dataset creation.
- Developed data validation checks and automated listings for safety and efficacy using R and Python.
- Conducted oncology-focused data analysis and authored reports for clinical publications.
- Developed automated algorithms for RECIST 1.1-based tumor response evaluation in R.
- Safety Data Validation: Designed automated checks for AE, CM, and Lab listings using R.
- Statistical Reporting: Created TLFs (Tables, Listings, and Figures) for Phase II Oncology studies.
- validation-Generated a randomization list (independent statistician) and supplied sealed envelopes to the sites for blind-breaking procedures.

## Senior Statistical Analyst Biostatistics

Bangalore

12 June 2023 -> 5 April 2024

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- Led biostatistics team (2 members) with daily scrums, code reviews, and statistical programming oversight for clinical research analyses
- Developed SAP tables/shells for Phase IV trials (survival, KM, Cox regression) and automated TLF generation
- Analyzed real-world survival outcomes (OS/PFS) using Kaplan-Meier, log-rank tests, and Cox regression with SAS/SQL
- Built LLM-based classification models and transcription tools using OpenAI (GPT-3.5, Whisper), Llama2, and DALL-E in R/Python
- Defined patient cohorts, integrated clinical trial and RWE data, and performed data cleaning with missing value/outlier handling
- Created KM curves, survival plots, and TLF summaries with clinical implications and data-driven recommendations
- Tools: SAS, R, Python, Bash, RMarkdown, RShiny, Git, Jira, DBeaver, RStudio, ChatGPT, Bard

## Associate Manager Clinical Data Analyst

Bangalore

14 October 2021 -> 9 June 2023

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- Trained team members on portfolio tasks, led studies in neuroscience/autoimmune therapeutic areas during trial phases
- Automated TLF generation from ADaM/SDTM using R, implemented Power BI dashboards for test data reporting with real-time validation
- Developed patient prediction models, Bayesian risk models (RBM), and outlier detection methods for site flagging
- Designed SAPs, collaborated with data management on Veeva EDC structure, CRF standards, and terminologies using R/CQL/SAS
- Supported FDA/MHRA submissions, ICH-GCP compliance, and regulatory inspections with clean patient trackers for study closeout
- Tools: Veeva EDC, JReview, SAS, RMarkdown, RShiny, Power BI, Power App, Git, RStudio, ChatGPT, Bard

## Clinical Data Analyst / Risk Analyst / Centralized Monitoring

Bangalore

17 Feb 2020 -> 13 Oct 2021

IQVIA

- Trained team in Spotfire dashboards and R Shiny automation for clinical research decision-making (Oncology, Neuroscience, Autoimmune)
- Developed R-based simulation models reducing analysis time by 40% across 5+ clinical trials
- Flagged site outliers using Bayesian modeling, Cook distance, Mahalanobis distance, and KNN in R
- Built RBM vs Non-RBM comparison dashboards using KRIs, developed novel R Shiny methodologies for IP discontinuation
- Coordinated with CROs/sponsors on data analytic plans, KRIs, and regulatory submissions (ICH-GCP certified)
- Tools: R, RShiny, Spotfire, Veeva EDC, MediRAVE, JReview, RMarkdown, Git, CTMS, ETMF

## Technical Consultant Statistician Data Analyst

Bangalore

5 June 2017 -> 14 Feb 2020

SPSS SOUTH ASIA PRIVATE LIMITED

- Supervised Predictive Analytics team developing time series and survival analysis algorithms with R validation
- Built simulation models using proprietary/vendor tools, applied ML (pattern recognition, visualization) to solve commercial problems
- Led data analytics lifecycle: extraction, transformation, quality control, exploratory analysis, and model deployment
- Created technical reports, presentations, tutorials, documentation, and case studies for client delivery
- Tools: IBM SPSS (Statistics, Modeler, Amos, Text Analytics), R, RShiny, Python, Tableau, Git

## Pre-sales/Post-sales Technical Associate Statistician

Bangalore

16 May 2016 -> 28 April 2017

STARCOM INFORMATION TECHNOLOGY LIMITED

- Supervised StarBI development team for time series algorithm development, R integration, and output validation
- Created business presentations, technical documents, case studies; maintained client relations and implemented promotional activities
- Tools: SPSS, SYSTAT, Sigmaplot, Sigmastat, TableCurve 2D/3D, Peakfit, R, RShiny, Python, Tableau, Git